```
}
}
```

Slip 23

1) Write a java program to calculate area of Cylinder and Circle.(Use super keyword)

```
class Circle{
double r;
Circle(double r){
this.r=r;
}
double area(){
return 3.142*r*r;
}
}
class Cylinder extends Circle{
double h;
Cylinder(double r,double h){
super(r);
this.h=h;
}
double area(){
return 2*3.142*r*(r+h);
}
}
public class AreaDemo{
public static void main(String[] args){
Circle c=new Circle(5);
Cylinder cy=new Cylinder(5,10);
```

```
System.out.println("Area of Circle: "+c.area());
System.out.println("Area of Cylinder: "+cy.area());
}
}
```

2) Write a program to sort HashMap by keys and display the details before sorting and after sorting.

```
import java.util.*;
public class SortHashMap{
public static void main(String[] args){
HashMap<Integer,String> map=new HashMap<>();
map.put(3,"C");
map.put(1,"A");
map.put(5,"E");
map.put(2,"B");
map.put(4,"D");
System.out.println("Before sorting: "+map);
TreeMap<Integer,String> sortedMap=new TreeMap<>(map);
System.out.println("After sorting: "+sortedMap);
}
}
```

OR

2) Write a java program to create using Runnable interface to display traffic signal.

```
class TrafficSignal implements Runnable{
public void run(){
try{
while(true){
System.out.println("RED - STOP");
Thread.sleep(2000);
```